

6. Limpets (*Patella vulgata*) are animals without backbones which live on rocky seashores, feeding on plants.



Photograph of limpets

- (a) State the scientific term for animals which do not have backbones.

[1]

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- (b) Students investigated the density of limpets (number per m^2) on two rocky shores in Anglesey. One shore was sheltered and one was exposed to heavy wave action.

seaweed



A sheltered shore



bare
rocks

An exposed shore

Students' method:

- Select a section on each shore of 300 m^2 .
- Place quadrats, each of area 1 m^2 , at 10 random co-ordinates in each of the two shore sections and collect data.
- Compare the data for the two shores.



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(i) **Table 6.1** shows one part of the students' risk assessment for the investigation.

Complete Table 6.1. [1]

Table 6.1

Hazard	Risk	Control measure
Sharp edges on rocks		

(ii) Describe in detail the techniques the students should use to **place their quadrats at random** and **collect data**. [3]

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The results of the investigation are shown in **Tables 6.2 and 6.3.**

Table 6.2

Number of limpets on exposed shore:

Quadrat number	1	2	3	4	5	6	7	8	9	10	Mean number per m ²	Estimated total number in the 300 m ² section of shore
Number of limpets	26	21	22	18	5	21	17	23	19	26	19.8	5940

Table 6.3

Number of limpets on sheltered shore:

Quadrat number	1	2	3	4	5	6	7	8	9	10	Mean number per m ²	Estimated total number in the 300 m ² section of shore
Number of limpets	30	22	26	31	28	25	23	19	31	26		

(iii) **Complete Table 6.3 for the sheltered shore** by calculating: [3]

- I. The mean number of limpets per m².
- II. The estimated total number of limpets in the 300 m² section of the shore.

Space for working



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(iv) I. From these results, state what the students could conclude about the density of limpets when they compared the two shores. [1]

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II. Suggest an explanation for this observation. [1]

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(v) State which of the quadrats (1 – 10) from the exposed shore shown in **Table 6.2** had an anomalous result and describe what should have been done to take account of this. [2]

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